Support Java Developer - Technical Assessment Task

Overview

This is a pre-screening technical assessment for Support Java Developer candidates.

Objective

Build a simple REST API for a product catalog management system using Spring Boot and Liquibase. Below you can find requirements, but we encourage you to be creative and make design decisions that you think are best for the problem at hand.

Business Context

You are building the backend API for an e-commerce platform that sells products from multiple producers. The system needs to handle products with varying complexity - some products may have just a few attributes (description, price), while others may have 50-200 attributes (dimensions, specifications, certifications, etc.). The platform currently works with approximately 100 different producers. For example we can have producer Samsung which sells TVs, phones, and laptops, and each of these product types has different sets of attributes.

Technical Requirements

Technology Stack

- **Java:** 21 or higher
- **Spring Boot:** 3.x
- **Liquibase:** for database schema management
- **Database:** H2 (in-memory) or PostgreSQL or Oracle (your choice)
- **Build Tool:** Maven

Business Requirements

The system must handle:

1. **Producers** - The platform works with approximately 100 different producers/manufacturers
2. **Products** - Each product belongs to one producer, but producers can have multiple products
3. **Product Attributes** - Products have varying numbers of attributes depending on their type and complexity:

- Some products may have just a few attributes (name, description, price)
- Others may have 50-200 attributes (dimensions, specifications, technical details, certifications, etc.)
- Example attributes might include: description, width, height, depth, weight, color, material, production_country, brand, warranty_period, energy_rating, certifications, and many more

Required API Functionality

Implement REST endpoints that support the following operations:

1. List Products

- Retrieve all products in the system
- Include relevant product information and producer details

2. Create Product

- Add new products to the system
- Associate products with producers

3. Update Product

- Modify existing product information
- Update product attributes

4. Delete Product

- Remove products from the system

Optional (Bonus)

- Producer-related endpoints
- Product filtering or search capabilities

What We're NOT Looking For

To keep the task focused, please **DO NOT** implement: - Authentication/Authorization (Spring Security) - Advanced network configuration - Caching layers - Message queues - Frontend interface - Deployment configurations (Docker, Kubernetes) - Comprehensive error handling framework (basic validation is sufficient)

Submission Guidelines

1. Publish your solution to your personal GitHub repository

2. Ensure the repository is publicly accessible or invite us as collaborators
3. Send the repository link via email to [CONTACT_EMAIL]
4. Include a README with setup and running instructions ---

Good luck! We look forward to reviewing your solution.